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RESEARCH ARTICLE

Graph-Aware Multimodal Deep Learning for Classification of Diabetic Retinopathy Images

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ABSTRACT Diabetic Retinopathy (DR) is one of the leading causes of blindness worldwide, where early detection is critical to preventing irreversible vision loss. Traditional diagnostic methods primarily rely on single-modality data, such as retinal images, or the analysis of image features, which may limit diagnostic accuracy. To overcome these limitations, we propose Diabetic Retinopathy Diagnosis, namely ''DRdiag'', a novel model proposed for the early diagnosis of DR. DRdiag integrates multiple modalities: the Fundus images and the duration of disease evolution which is an important factor in the diagnosis of DR as the duration of the disease directly influences the onset and progression of retinal lesions, leveraging two distinct models: a Convolutional Neural Network (CNN) based on DenseNet121 for image feature extraction and a Graph Neural Network (GNN) for capturing complex relationships between patient features. By combining the strengths of these advanced models, DRDiag aims to provide a more comprehensive and accurate diagnostic system, ultimately supporting timely intervention and improved patient outcomes. The experiments are conducted using two datasets, APTOS2019 and Messidor-2, which provide diverse retinal images for robust evaluation. The proposed model achieves high performance on diabetic retinopathy classification. On Messidor-2, it attains 0.976 accuracy, 0.957 kappa, 0.961 F1-score, 0.958 recall, 0.965 precision, and 0.990 specificity. On APTOS2019, it achieves 0.980 accuracy, 0.960 kappa, 0.963 F1-score, 0.959 recall, 0.968 precision, and 0.995 specificity. These results confirm its robustness and reliability.

INDEX TERMS Classification, convolutional neural network (CNN), diabetes, diabetic retinopathy (DR), graph neural network (GNN).

NOMENCLATURE

AI	Artificial Intelligence .
APTOS	Asia Pacific Tele-Ophthalmology Society.
CAD	Computer-Assisted Diagnostic.
CNN	Convolutional Neural Network.
DDR	Dataset for Diabetic Retinopathy.
DGCN	Deep Graph Convolutional Network.
DME	Diabetic Macular Edema.
DR	Diabetic Retinopathy.
ECOC	Error Correction Output Code.
FCN	Fully Convolutional Network.
GCN	Graph Convolutional Networks.
GNN	Graph Neural Network.
ICDR	International Clinical Diabetic Retinopathy.
IDRiD	Indian diabetic retinopathy image dataset.
IoU	Intersection over Union.
LSTM	Long Short Term Memory.
MMDA	Multi-Model Domain Adaptation.
Optuna	A framework for Bayesian optimization used to tune hyperparameters.
rDR	referable Diabetic Retinopathy.
ROI	Region of Interest.
SSGL	Semi-Supervised Graph Learning.
VAE	Variational Autoencoder.

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I. INTRODUCTION

Artificial Intelligence in medical imaging enhances diagnosis across various fields: ophthalmology such as diabetic retinopathy and glaucoma [1], radiology such as tumors and fractures [2], cardiology such as heart disease detection [3], neurology such as stroke and brain tumors [4], and dermatology such as skin cancer [5]. By improving accuracy, speed, and interpretability, AI is revolutionizing healthcare diagnostics. In this paper, we focus on eye diseases, leveraging AI to improve the detection and diagnosis of conditions like diabetic retinopathy and glaucoma. Diabetic retinopathy, a serious complication of diabetes mellitus, arises from prolonged periods of uncontrolled high blood sugar, which gradually damages the blood vessels in the retina. If not treated, this condition can cause significant vision loss and potentially lead to blindness [6]. The growing burden of diabetic retinopathy highlights the need for innovative solutions to improve early detection and treatment. Leveraging advancements in medical artificial intelligence (AI), recent researchers [7], [8], [9], [10], [11] are now exploring novel techniques as a promising approach to address this challenge. Detecting diabetic maculopathy manually requires significant time and effort from ophthalmologists. To enhance the analysis of retinal image structures, advanced image processing techniques are essential for developing effective Computer-Assisted Diagnostic (CAD) systems [12]. Furthermore, early detection of diabetic retinopathy is crucial for individuals with diabetes to reduce the risk of vision loss. By integrating powerful computational methods with clinical data, AI offers the potential to revolutionize the diagnosis and management of diabetic retinopathy, ensuring timely intervention and better outcomes for patients. Medical AI has made remarkable advancements in recent years, driven by the rapid development of deep learning techniques. For example, deep neural networks have demonstrated accuracy on par with, or even exceeding, that of clinical experts in tasks such as referral recommendations for sight-threatening retinal conditions and the detection of pathologies [13]. Existing DR classification methods primarily analyze retinal images without leveraging patient-specific information, limiting their predictive capabilities. This work aims to improve DR classification by incorporating both image-based and patient-specific clinical features.

This paper introduces a multimodal model called "DRdiag" for diabetic retinopathy (DR) image classification that combines Graph Neural Networks (GNNs) and Convolutional Neural Networks (CNNs). The model leverages both the intrinsic features of DR images and the progression duration of diabetes. Specifically, the CNN extracts deep image features, while a two-layer GNN captures complex relationships among patient attributes, incorporating the duration of their diabetic condition.

The rest of the paper is structured as follows. Section II, after introducing DR, reviews the related works for diabetic retinopathy discovery. Section III describes the proposed model. Section IV presents experimental settings with the dataset explanation and compares our proposal with other state-of-the-art methods. Finally, Section VI concludes the paper and presents future works.

II. BACKGROUND AND RELATED WORK

In this section, we will introduce diabetic retinopathy and some related works on DR detection.

A. DIABETIC RETINOPATHY

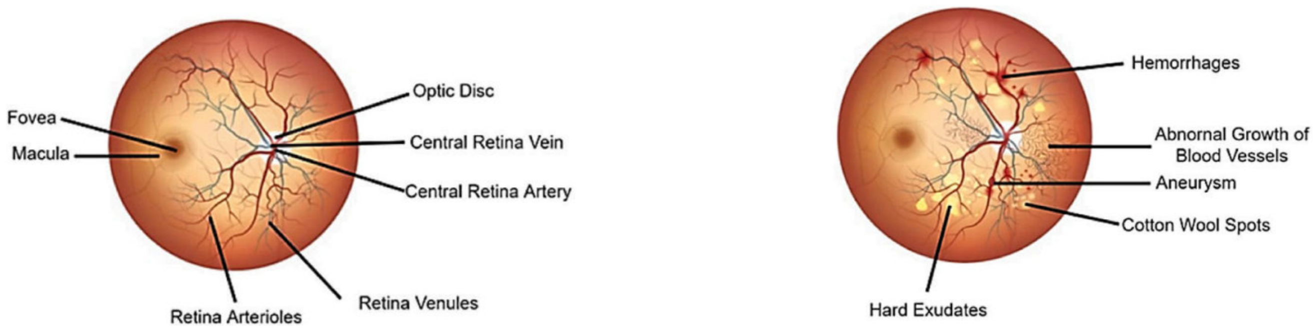
Diabetic retinopathy is a serious condition commonly resulting from prolonged diabetes. Early detection is crucial, as delayed or ineffective treatment can lead to vision loss. Diabetes itself is a significant disease characterized by the body's inability to regulate blood sugar levels, leading to elevated glucose levels in the bloodstream. This condition can result in severe complications, including kidney failure, heart attacks, nerve damage, and vision impairment. In 2022, it was reported that diabetes directly affected approximately 9% of the overall global populace. By 2030, the number of people suffering from diabetes is expected to increase to around 12% [14]. Ophthalmologists typically diagnose DR progression from non-proliferative DR (NPDR), marked by hard exudates and edema, to proliferative DR (PDR), where abnormal blood vessels grow, risking severe vision loss. Retinal hemorrhages signal worsening DR, while exudates result from fluid leakage. Microaneurysms, early DR signs, appear as small red specks and can lead to retinal damage if untreated. Figure 1 illustrates a comparison of retina conditions, highlighting the differences between a healthy retina (Figure 1a) and one affected by diabetic retinopathy (Figure 1b).

Research on DR screening and commercial diagnostic devices often focuses on the binary classification task of referable DR (rDR). This classification defines the positive class as either moderate or more severe stages on the International Clinical Diabetic Retinopathy (ICDR) scale or the presence of diabetic macular edema (DME) [15]. Table 1 summarizes the ICDR stages and their observable manifestations during dilated ophthalmoscopy.

Figure 2 highlights the correlation between the duration of diabetes and diabetic retinopathy (DR). Effectively tackling this issue necessitates a multidisciplinary approach, integrating healthcare professionals, cutting-edge diagnostic technologies, and comprehensive patient education [16].

B. RELATED WORKS ON DIABETIC RETINOPATHY DETECTION

A significant amount of research has been carried out in the literature to develop advanced methods for the detection and classification of diabetic retinopathy. These studies have explored various approaches, including traditional machine learning techniques, deep learning models, and hybrid



(a) Healthy retina.

(b) Diabetic retinopathy.

FIGURE 1. Comparison of retina conditions: (a) healthy and normal retina, and (b) retina affected by diabetic retinopathy, showing damaged blood vessels and other abnormalities [14].

TABLE 1. International Clinical Diabetic Retinopathy (ICDR) scale and observable manifestations.

Disease Level	Observable Manifestations on Dilated Ophthalmoscopy
(0) No retinopathy	No abnormalities detected.
(1) Mild nonproliferative DR	Presence of microaneurysms only.
(2) Moderate nonproliferative DR	Lesions more advanced than microaneurysms but not severe nonproliferative DR.
(3) Severe nonproliferative DR	Any of the following: • More than 20 intraretinal hemorrhages in each of 4 quadrants. • Definite venous beading in 2 or more quadrants. • Prominent intraretinal microvascular abnormalities in 1 or more quadrants, without signs of proliferative DR.
(4) Proliferative DR	One or more of the following: • Neovascularization. • Vitreous or preretinal hemorrhage.

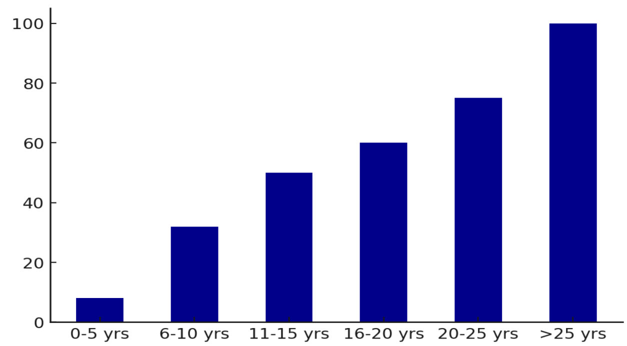


FIGURE 2. DR association with the duration of diabetes in patients [16].

methodologies, to enhance the accuracy and reliability of DR diagnosis.

The study in [17] employs capsule networks to classify diabetic retinopathy stages from the Messidor dataset fundus images. The proposed model achieves an accuracy of 97.98%, demonstrating superior performance compared to traditional convolutional neural networks.

In [18], the authors propose a Multi-Model Domain Adaptation (MMDA) framework to classify DR images without labeled target data. The methodology (1) Leverages multiple pre-trained source models (DDR, IDRiD, Messidor, Messidor-2) and a weighted pseudo-labeling mechanism and (2) Adapts knowledge to a target dataset (APTOS2019) through a model weight determination module and an information maximization loss function. Extensive experiments showed competitive performance, highlighting the efficacy of MMDA in transferring knowledge without requiring source datasets.

A method based on CNN and GNN for DR grading was proposed in [19]. Key aspects include: (1) Extracting deep features using CNN and modeling relationships among these features with GNN and (2) Fusing CNN and GNN outputs using adaptive weights for final classification. The approach demonstrated improved accuracy and F1 scores on the APTOS2019 and Messidor-2 datasets, emphasizing the importance of relational feature learning for distinguishing overlapping DR severity levels.

The authors of [12] introduce a hybrid model combining Variational Autoencoder (VAE) and Graph Convolutional Networks (GCN) to classify DR severity levels. The methodology involves (1) Using Fully Convolutional Networks (FCN) for Region of Interest (ROI) extraction, (2) Employing VAE for feature encoding and GCN for topological learning, and (3) Modeling retinal images as graphs, where each pixel is a node. The model was validated on the Kaggle and EyePACS datasets, demonstrating superior performance in accuracy and specificity compared to CNN-based approaches. Key challenges addressed include enhancing interpretability and optimizing feature representations.

The paper in [20] presents a transfer learning approach leveraging ShuffleNet and ResNet-18 models with features extracted from the average pooling layers and fed to an Error Correction Output Code (ECOC) ensemble. Using datasets like APTOS and EyePac+Messidor-2, the method achieves 96% accuracy for binary DR grading and 82% for multi-class grading. Adaptive differential evolution optimizes parameters, enhancing detection accuracy while addressing class imbalance.

In [21], a robust pipeline integrating preprocessing, DR-UNet with atrous spatial pyramid pooling, and an attention-aware DCNN was proposed. The model achieved 99.2% classification accuracy and 87.1% IoU (Intersection over Union) in segmenting lesions, highlighting its efficacy in early DR detection.

Talukder et al. [22] propose a machine learning approach for diabetic retinopathy detection using transfer and ensemble learning. They evaluated nine pre-trained models, including DenseNet121 and DenseNet169, which achieved 100% accuracy through ensemble techniques. Their work highlights data augmentation, dropout layers, and ensemble averaging for performance optimization. Validation on combined IDRiD and Mendeley datasets demonstrated robust results for early detection and classification.

Authors in [23] introduce a Semi-Supervised Graph Learning (SSGL) algorithm for diabetic retinopathy detection, leveraging labeled and unlabeled data relationships. The SSGL framework improves accuracy, specificity, and sensitivity while addressing data imbalance. It utilizes advanced augmentation techniques and personalized patient risk prediction based on demographic and medical data. Testing on two public datasets showed superior performance against benchmarks.

Luo et al. [24] developed a CNN incorporating both local and long-range dependencies for diabetic retinopathy detection. The method uses patch-based relationships to enhance feature extraction, addressing lesions' spatial variability. The approach achieves higher accuracy compared to existing methods on Messidor and EyePACS datasets, highlighting the flexibility of embedding long-range units into trained models.

In another research [25], the authors propose a novel hybrid model for DR detection and grading. The authors address imbalanced datasets by applying the SMOTE algorithm to balance the instances across DR severity grades. A Deep Graph Convolutional Network (DGCN) is then utilized to extract class-specific features by leveraging graph-based relationships. These features are classified using an XGBoost classifier, yielding high accuracy, sensitivity, and specificity. The model's performance surpasses existing methods when tested on the EyePACS and Messidor datasets. This work highlights the potential of combining graph learning and ensemble classifiers for efficient and accurate DR grading.

In [26], the authors introduce DRNet13, a CNN model that achieves 97% accuracy in diabetic retinopathy detection using fundus images. The model utilizes enhanced preprocessing, including median filtering and gamma correction, and expands the dataset through augmentation. DRNet13 surpasses 15 pre-trained models in speed and efficiency, demonstrating its potential for automated, early-stage DR diagnosis.

Authors in [27] retrain AlexNet for automated detection of glaucoma and diabetic retinopathy using retinal fundus images. Validation accuracies of AlexNet, ResNet50, and GoogLeNet range from 89.7% to 94.3%, emphasizing the feasibility of CNNs for accessible, early disease detection. Grad-CAM visualizations enhance the interpretability of model predictions.

The paper [28] proposes an Explainable AI (XAI)-based model for diagnosing Diabetic Retinopathy (DR), improving interpretability and accuracy in AI-driven healthcare. Using CNNs and XAI techniques like LIME, the model achieves 94% accuracy, providing transparent and reliable explanations for predictions. It enhances diagnostic confidence by highlighting key retinal features that influence decisions.

Most existing approaches for diabetic retinopathy (DR) classification rely exclusively on retinal images, neglecting valuable contextual information that could significantly enhance diagnostic accuracy. Additionally, these methods fail to capture relationships between patients, which could provide critical insights into shared risk factors and disease progression patterns.

To address these limitations, we propose a novel multimodal model that integrates image-based features extracted using Convolutional Neural Networks (CNNs) with patient attribute relationships modeled through Graph Neural Networks (GNNs). Unlike prior studies, our approach incorporates patient specific clinical data, particularly the duration of diabetes which is a key factor influencing DR onset and progression. By combining visual features with this essential contextual information and leveraging GNNs to model complex patient relationships, our method provides a more comprehensive and accurate DR classification system, surpassing traditional image-only approaches.

III. THE PROPOSED APPROACH

DRdiag combines CNN and GNN to leverage both the intrinsic features of DR images and the duration of diabetic disease progression. Specifically, the model employs CNN to extract deep image features and utilizes a two-layer GNN to capture complex relationships between patient attributes, considering the duration of their diabetic condition. Finally, an adaptive weight mechanism fuses the outputs of the two networks for robust classification. Figure 3 presents the architecture of the proposed approach for the classification of diabetic retinopathy.

The subsections below will detail the proposed architecture.

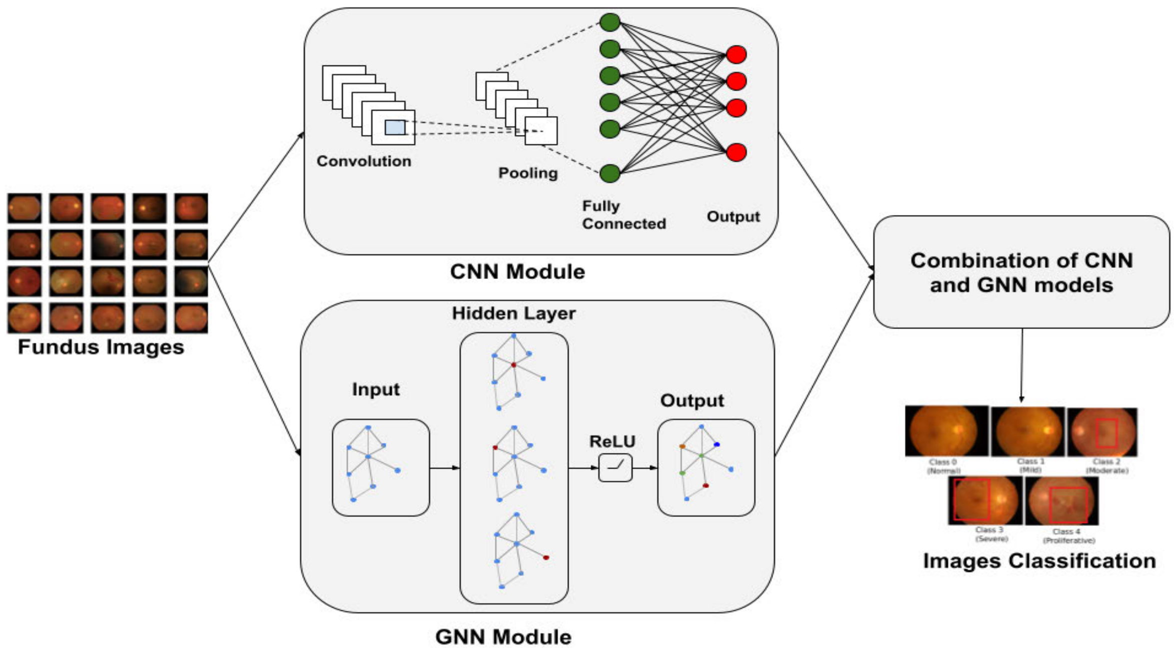


FIGURE 3. The architecture of DRdiag model.

A. DATA PREPROCESSING

Specifically, we resized all retinal images to 224×224 pixels and normalized them. Additionally, we applied standard augmentation methods to improve generalization, including random rotations ($\pm 15°$), horizontal flipping, contrast adjustments, and slight Gaussian blurring. These techniques help mitigate overfitting and improve the model's ability to handle variations in real-world scenarios.

B. FEATURE EXTRACTION USING CNN

DenseNet CNN has a pattern where each layer is connected to every other in a feed-forward manner. This enables networks to grow deeper. This is the connectivity structure known in DenseNet, as every layer takes input from all its preceding layers and passes on their feature maps to subsequent levels. The dense connection facilitates direct information flow across the network and supports feature reuse throughout the (layer stack) [29], [30]. DenseNet121 was chosen as the backbone CNN due to its superior performance. As shown in this work [31], the proposed CNN with DenseNet achieved the highest validation accuracy (96.22%), outperforming ResNet (93.18%), E-DenseNet (91.6%), LSTM-based CNN (90%), and AdaBoost (88.21%). Additionally, its low validation loss (0.1142) indicates better convergence and efficiency in error reduction. The CNN-based feature extractor is implemented as follows: Base Model, Global Average Pooling, and Fully Connected Layer.

The base model used is DenseNet121, pre-trained on the ImageNet dataset (`IMAGENET1K_V1`) [32]. This model extracts deep features from input images. The key steps include:

- A feature extraction step features($\mathbf{x}$), projecting the images into high-dimensional feature spaces;
- an adaptation layer to adjust feature dimensions for the target task.

The feature map $\mathbf{F} \in \mathbb{R}^{B\times C\times H\times W}$ - where B is the batch size, C is the number of channels, and H, W are spatial dimensions - is reduced using global average pooling:s

$$\mathbf{F}_{\text{pooled}} \in \mathbb{R}^{B\times C}$$

This reduces the spatial dimensions (H, W) into a single vector for each channel using (equation 1):

$$\mathbf{F}_{\text{pooled}}[b, c] = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} \mathbf{F}[b, c, h, w] \tag{1}$$

The reduced feature vector is passed through a fully connected (FC) layer (equation 2):

$$\mathbf{X} = \mathbf{F}_{\text{pooled}} \mathbf{W}_{\text{fc}} + \mathbf{b}_{\text{fc}} \tag{2}$$

where:

- $\mathbf{W}_{\text{fc}} \in \mathbb{R}^{C\times d_{\text{out}}}$ is the weight matrix of the FC layer.
- $\mathbf{b}_{\text{fc}} \in \mathbb{R}^{d_{\text{out}}}$ is the bias vector.
- $\mathbf{X} \in \mathbb{R}^{B\times d_{\text{out}}}$ represents the final output of the CNN feature extractor.

The CNN feature extractor is defined as follows:

1) The input image $\mathbf{x}$ is passed through the feature extractor (equation 3):

$$\mathbf{F} = \text{features}(\mathbf{x}) \tag{3}$$

2) Global average pooling is applied (equation 4):

$$\mathbf{F}_{\text{pooled}} = \text{GAP}(\mathbf{F}) \tag{4}$$

3) The reduced vector is fed into the FC layer (equation 5):

$$\mathbf{X} = \mathbf{F}_{\text{pooled}}\mathbf{W}_{\text{fc}} + \mathbf{b}_{\text{fc}} \tag{5}$$

C. GNN MODULE

Graph Neural Networks have recently achieved significant advancements in various practical domains, such as drug discovery, medical image analysis, and recommendation systems. When applied to DR detection, the graph-based structure of GNNs leverages both labeled and unlabeled data. By efficiently capturing data relationships, handling unlabeled data, and enabling scalability, GNNs enhance diagnostic accuracy for DR [25].

Our GNN module is based on a **Graph Convolutional Network (GCN)**, which captures complex relationships between patient attributes. The GCN model enables information propagation across patients, leveraging shared clinical characteristics to enhance disease prediction.

1) GRAPH STRUCTURE

The graph is constructed where:

- **Nodes** represent individual patients.
- **Edges** represent relationships between patients, defined based on disease progression similarity.

2) EDGE RELATIONSHIPS

Edges between patients are established using a **similarity metric** that takes into account shared clinical attributes, including:

- **Duration of diabetes**: The number of years since diagnosis.

Patients with **similar disease progression trends** have stronger connections, allowing the model to learn meaningful representations from neighboring nodes.

The main idea of the described GNN model is to iteratively update node representations by leveraging both the graph structure and the node features. At each layer, the model applies a transformation that combines information from neighboring nodes using the normalized adjacency matrix $\hat{\mathbf{A}}$ and learns feature transformations through weight matrices $\mathbf{W}$.

In the first layer, the input features $\mathbf{X}$ are transformed into a hidden representation $\mathbf{H}^{(1)}$ according to the following equation 6:

$$\mathbf{H}^{(1)} = \text{ReLU}(\hat{\mathbf{A}}\mathbf{X}\mathbf{W}^{(0)}) \tag{6}$$

where $\hat{\mathbf{A}}$ ensures smooth aggregation across neighboring nodes, and $\mathbf{W}^{(0)}$ learns relationships between features.

The second layer further refines the node embeddings by propagating the updated node features $\mathbf{H}^{(1)}$ through another transformation (equation 7):

$$\mathbf{H}^{(2)} = \hat{\mathbf{A}}\mathbf{H}^{(1)}\mathbf{W}^{(1)} \tag{7}$$

producing the final node representations $\mathbf{H}^{(2)}$. This approach effectively captures both the structural and feature information of the graph.

D. COMBINATION OF CNN AND GNN MODELS

Feng et al. [19] conducted an ablation experiment to demonstrate the impact of fusing GNN with CNN outputs. The results indicate that integrating GNN with CNN significantly improves classification performance compared to using a CNN model alone.

The proposed model combines CNNs and GNNs to classify diabetic retinopathy using both image data and graph-structured data. The architecture integrates CNN and GNN outputs through an adaptive fusion mechanism to leverage the complementary strengths of both modalities.

The model consists of the following components:

- **CNN Feature Extractor:** A convolutional neural network extracts features from input images, producing a feature vector $\mathbf{f}_{\text{cnn}}$ of dimension 1024. These features are passed through a fully connected layer, $\mathbf{fc}_{\text{cnn}}$, to output class scores $\mathbf{s}_{\text{cnn}}$.
- **GNN Module:** A graph neural network processes node features and graph structure. Node features are enriched by concatenating a duration vector, resulting in input features $\mathbf{X}_{\text{gnn}} \in \mathbb{R}^{N\times(d_{\text{cnn}}+1)}$. The GNN outputs aggregated graph features, which are passed through another fully connected layer, $\mathbf{fc}_{\text{gnn}}$, to generate class scores $\mathbf{s}_{\text{gnn}}$.
- **Adaptive Fusion:** The outputs of the CNN and GNN are combined using an adaptive weight parameter $\alpha \in [0, 1]$, initialized to 0.5 and learned during training. The final scores are computed as [19] (equation 8):

$$\mathbf{s}_{\text{final}} = (1-\alpha)\mathbf{s}_{\text{cnn}} + \alpha\mathbf{s}_{\text{gnn}}. \tag{8}$$

IV. EXPERIMENTS

This Section will show the experiments that have been set and executed to prove the effectiveness of our approach. More in particular, in the following subsections, we will introduce the used datasets, detail the chosen settings, and show the obtained results.

A. DATASET

The APTOS2019 and MESSIDOR-2 datasets [33], [34] were used in order to evaluate the performance of our proposed model. Such datasets are better detailed in the following.

1) APTOS 2019

The APTOS dataset [33], from the Asia Pacific TeleOphthalmology Society (APTOS), includes high-quality retinal images and is used for the detection and grading of diabetic retinopathy. The dataset has been collected by the Aravind Eye Hospital in India. It features a diverse set of images, which are crucial for training models to handle various lighting conditions, camera models, and ethnicities. The dataset consists of 3,662 images divided into five classes of which there are 1,805 fundus images for healthy eyes, 370 images for mild NPDR (Nonproliferative Diabetic Retinopathy), 999 images for moderate NPDR, 193 images for severe NPDR, and 295 images for PDR (Proliferative Diabetic Retinopathy).

2) MESSIDOR-2

The MESSIDOR-2 dataset [34] has been publicly distributed since 2008. It is of interest mainly to researchers in a relatively specialized domain: retinal image processing and, more specifically, computer-assisted diagnosis of diabetic retinopathy. It contains 1,748 retinal images of 874 subjects. The images were captured using 8 bits per color plane at 1440*960, 2240*1488, or 2304*1536 pixels.

B. EVALUATION SETTINGS

Through extensive experimentation, we carefully tuned the hyperparameters to achieve optimal performance, balancing convergence speed, stability, and generalization while minimizing overfitting. We employed a systematic validation process, including k-fold cross-validation (with k = 5) and Bayesian optimization (via Optuna), to identify the best hyperparameters for our models.

For the APTOS2019 dataset, we explored various learning rates, including 0.0005, 0.0002, 0.0001, 0.005, 0.002, 0.001, and 0.01. After extensive testing, we selected a learning rate of 0.001, as it provided the best trade-off between convergence speed and stability. The number of epochs was systematically evaluated within a range of 1 to 100, and we determined that 28 epochs achieved the best performance without overfitting. Additionally, batch sizes of 16, 32, and 64 were tested, with 32 being chosen as the optimal size for balancing model generalization, training efficiency, and memory constraints.

For the MESSIDOR-2 dataset, the same learning rate values were explored, with 0.001 again proving to be the optimal choice. The number of epochs was tuned between 1 and 100, and we found that 30 epochs provided the best balance between performance and training stability. As with the APTOS2019 dataset, we tested batch sizes of 16, 32, and 64, selecting 32 as the ideal choice.

The hyperparameters tested, along with the chosen values, are described in detail in Table 2 and Table 3 for APTOS2019 and MESSIDOR-2, respectively.

TABLE 2. Hyperparameters for the APTOS2019 dataset.

Hyperparameter	Values Explored	Selected Values
Learning rate	0.0005, 0.0002, 0.0001, 0.005, 0.002, 0.001, 0.01	0.001
Number of epochs	1,2,..., 100	28
Batch size	16, 32, 64	32

TABLE 3. Hyperparameters for the MESSIDOR-2 dataset.

Hyperparameter	Values Explored	Selected Values
Learning rate	0.0005, 0.0002, 0.0001, 0.005, 0.002, 0.001, 0.01	0.001
Number of epochs	1,2,..., 100	30
Batch size	16, 32, 64	32

C. RESULTS OF THE PROPOSED MODEL

Six evaluation metrics: accuracy (Acc), kappa, F1-score (F1), Recall, Precision, and Specificity are employed to assess the performance of the proposed classification model. These metrics are defined as follows:

- **Accuracy (Acc):** Accuracy is the ratio of correctly classified instances to the total number of instances, defined as follows (Equation 9):
$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$
where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.
- **Kappa:** Kappa measures the agreement between predicted and actual classifications, adjusting for chance agreement (Equation 10):
$$\text{Kappa} = \frac{P_o - P_e}{1 - P_e} \tag{10}$$
where P_o is the observed agreement, and P_e is the expected agreement by chance.
- **F1-score (F1):** The F1-score is the harmonic mean of precision and recall, given by (Equation 11):
$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$
- **Precision:** Precision measures the proportion of predicted positive instances that are actually positive (Equation 12):
$$\text{Precision} = \frac{TP}{TP + FP} \tag{12}$$
- **Recall (Sensitivity):** Recall measures the proportion of actual positive instances that are correctly identified (Equation 13):
$$\text{Recall} = \frac{TP}{TP + FN} \tag{13}$$
- **Specificity:** Specificity measures the proportion of negative instances that are correctly identified (Equation 14):
$$\text{Specificity} = \frac{TN}{TN + FP} \tag{14}$$

Figures 4 and 5 present the results of the proposed model on the APTOS2019 and MESSIDOR-2 datasets, respectively.

The figures illustrate the performance of the proposed model on the APTOS2019 and MESSIDOR-2 datasets in terms of key classification metrics: Accuracy, Cohen's Kappa, Specificity, Precision, Recall, and F1-Score. The analysis spans 28 and 30 epochs, respectively, and demonstrates the effectiveness of the model in diabetic retinopathy classification. The datasets were divided into training, validation, and test sets to ensure optimal model training, hyperparameter tuning, and unbiased performance evaluation. The model exhibits a strong learning capability, as indicated by the progressive improvement in its evaluation metrics.

TABLE 4. Comparison of DRDiag with state-of-the-art methods.

Reference	Method(s)	Used dataset(s)	Obtained results
[19]	CNN (ResNet 50, DenseNet 121) + GCN	Messidor-2 APTOS2019	Accuracy = 0.680 Kappa = 0.673 F1 Score = 0.660 Accuracy = 0.855 Kappa = 0.906 F1 Score = 0.849
[15]	DINOv2 base vision transformer (ViT-base)	Messidor-2 APTOS	AUC = 0.976 F1 = 0.857 MC-ACC = 0.782 L-Kappa = 0.758 Q-Kappa = 0.865 AUC = 0.969 F1 = 0.894 MC-ACC = 0.749 L-Kappa = 0.799 Q-Kappa = 0.901
[35]	Vision Transformer	MESSIDOR APTOS	AUC = 0.973 Accuracy = 0.969 F1 Score = 0.947 Sensitivity = 0.941 Precision = 0.954 G-Mean = 0.947 AUC = 0.982 Accuracy = 0.897 wF1 = 0.881 wKappa = 0.923
[36]	Hybrid models (CNN, VGG16 and VGG19)	APTOS2019, Messidor-2 and Local public DR	Accuracy = 0.906 Recall = 0.950 Precision = 0.946 F1 Score = 0.940
[37]	Transfer Learning-based Hybrid GoogleNet and ResNet-18 features with SVM Classifier	APTOS	Accuracy = 0.892
[28]	Explainable AI based model	Diagnosis of Diabetic Retinopathy	Accuracy = 0.94
[38]	Swin Transformer	APTOS	Train Acc = 0.884 Train-loss = 0.669 top-2-acc = 0.956 Val-Acc = 0.698 Val-loss = 1.085 Valtop-2-acc = 0.797
[39]	Federated learning using Vision Transformer	MESSIDOR APTOS	Accuracy = 0.79 AUC = 0.83 Specificity = 0.83 Sensitivity = 0.55 Accuracy = 0.95 AUC = 0.95 Specificity = 0.95 Sensitivity = 0.95
DRDiag	Multimodal approach using CNN (DenseNet121) + GNN	MESSIDOR APTOS	**Accuracy = 0.976** **Kappa = 0.957** **F1 Score = 0.961** **wF1 = 0.973** **Recall = 0.958** **Precision = 0.965** **Specificity = 0.990** **Accuracy = 0.980** **Kappa = 0.960** **F1 Score = 0.963** **wF1 = 0.978** **Recall = 0.959** **Precision = 0.968** **Specificity = 0.995**

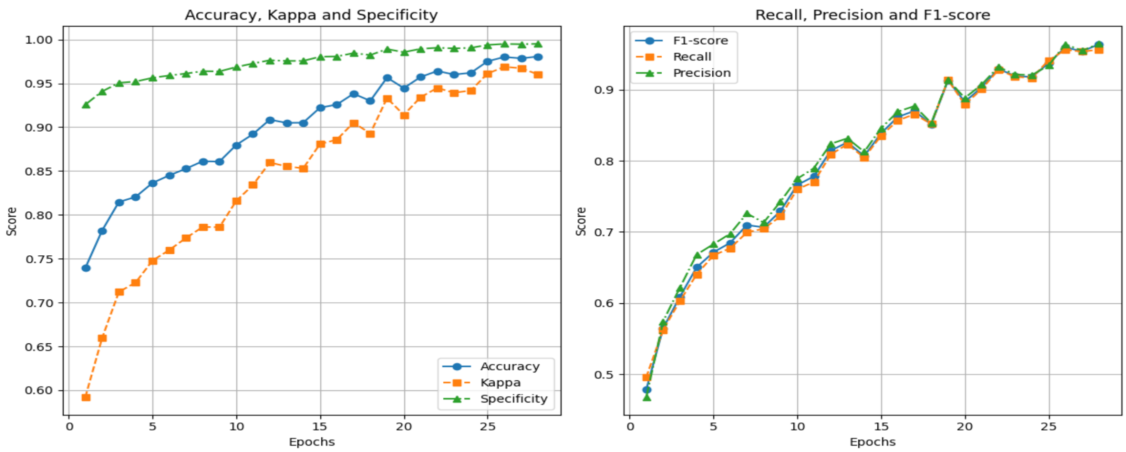


FIGURE 4. Performance metrics of the proposed model on the APTOS2019 Dataset per epoch.

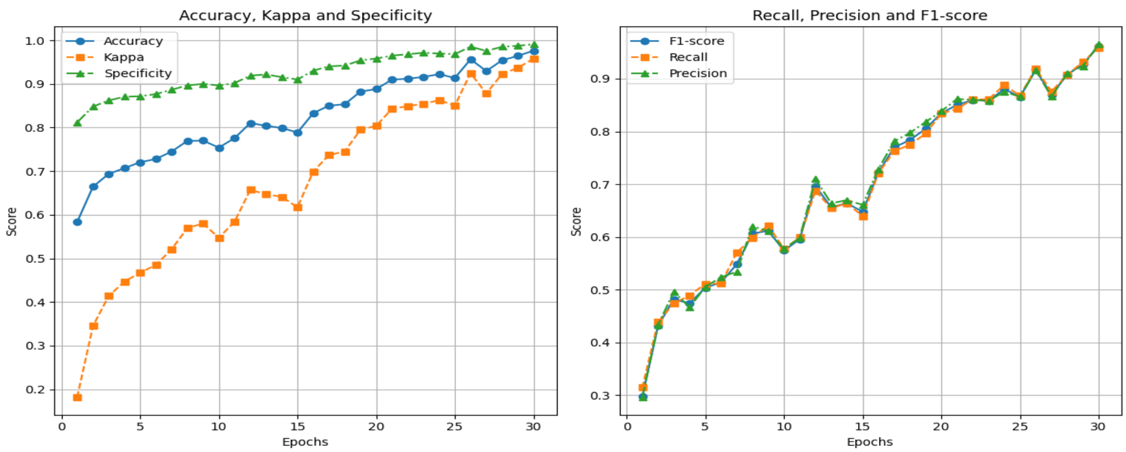


FIGURE 5. Performance metrics of the proposed model on MESSIDOR-2 Dataset per epoch.

- **Accuracy, Cohen's Kappa, and Specificity:** Accuracy exhibits a steady increase, starting at approximately 55-60% and surpassing 95% by the final epoch, demonstrating the model's ability to refine its predictions over time. Similarly, Cohen's Kappa score, which measures agreement beyond chance, rises from an initial value of around 0.2 to over 0.9, indicating improved prediction consistency. Specificity remains high throughout, beginning at approximately 85% and approaching near-perfect values (99%), suggesting that the model effectively distinguishes between diabetic and non-diabetic cases, minimizing false positives.
- **Precision, Recall, and F1-Score:** Precision remains consistently high, indicating that the model's positive predictions are reliable, exceeding 95% in the final epochs. Recall, which measures the model's ability to correctly identify positive cases, starts lower (around 30-50%) but steadily improves beyond 95%, showcasing enhanced sensitivity to diabetic retinopathy cases. The F1-Score, balancing Precision and Recall, shows a significant rise from 0.3-0.5 in the early epochs to over 0.95, confirming the model's robustness in handling both minority and majority classes.

These results indicate that the proposed model achieves high classification performance across multiple key metrics. The continuous improvement in Accuracy, Kappa, and Specificity confirms the model's ability to generalize effectively, while the near-perfect Precision, Recall, and F1-Score highlight its reliability in making accurate predictions. The model demonstrates strong robustness across different datasets, making it a promising tool for automated diabetic retinopathy detection.

D. COMPARISON WITH OTHER STATE-OF-THE-ART METHODS

The evaluation results presented in Table 4 confirm the superior performance of DRDiag compared to state-of-the-art approaches, including those by Feng et al. [19], Men et al. [15], Liu et al. [35], Benaouer et al. [36], Shahzad et al. [28], Butt et al. [37], Dihin et al. [38], and

Chetoui et al. [39]. The comparison is conducted on the MESSIDOR-2 and APTOS2019 datasets.

V. DISCUSSION

For MESSIDOR-2, DRDiag achieves the highest classification accuracy at 0.976, surpassing Feng et al.'s 0.680 and Liu et al. [35] 0.969. The F1-score of 0.961 also outperforms Liu et al. (0.947) and Men et al.'s (0.857). Furthermore, the kappa statistic of 0.957, which reflects inter-rater agreement, is the highest among all models. Compared to the Vision Transformer model, which achieved an accuracy of 0.969 and an AUC of 0.973, DRDiag shows a clear improvement. Additionally, in comparison to Chetoui et al. [39], whose Vision Transformer-based federated learning model reached an accuracy of 0.79 and an AUC of 0.83, DRDiag significantly enhances diagnostic performance, underscoring its robustness.

For APTOS2019, DRDiag maintains its superior performance, attaining an accuracy of 0.980 and an F1-score of 0.963, both exceeding Liu et al. [35], which reported an accuracy of 0.897 and a weighted F1-score of 0.881. Additionally, its kappa score of 0.960 surpasses competing models. Compared to the Vision Transformer model, which achieved an accuracy of 0.897 and an AUC of 0.982, DRDiag provides a significant improvement in classification accuracy. Furthermore, when compared to the Explainable AI-based model, which reached an accuracy of 0.94, DRDiag demonstrates superior classification performance. Relative to Benaouer et al. [36], whose hybrid CNN-based approach achieved an accuracy of 0.906, an F1-score of 0.940, and a recall of 0.950, DRDiag exhibits a substantial improvement. Additionally, while the Swin Transformer-based model by Dihin et al. [38] showed promising training results, including a top-2 accuracy of 0.956, DRDiag consistently outperforms other methods across multiple evaluation metrics.

Overall, DRDiag achieves state-of-the-art performance on both datasets, setting a new benchmark in diabetic retinopathy classification. These improvements highlight its potential for deployment in clinical settings, where precision and reliability are crucial for early diagnosis and treatment.

To assess the real-world applicability of DRDiag, we validated the model with the expertise of an ophthalmologist, who is also a co-author of this study. The ophthalmologist provided clinical insights and feedback on the model's predictions, ensuring its relevance and potential usefulness in practical medical settings. While this initial evaluation supports the model's applicability, further large-scale clinical validation with multiple medical professionals is planned for future work.

VI. CONCLUSION AND FUTURE WORK

Diabetic retinopathy is a disease of the eye caused by diabetes and is considered one of the leading causes of blindness among the working-age population worldwide. This condition, which involves damage to the blood vessels in the retina, progresses gradually and often goes unnoticed until significant vision impairment occurs [33]. Early detection is crucial to preventing diabetic retinopathy. However, conventional methods for this purpose are time-consuming, have low prediction accuracy, and often fail to detect the disease in its early stages. When the condition is identified at advanced stages, the likelihood of effective treatment and recovery significantly decreases. Hence, achieving early detection with high accuracy is essential in managing diabetic retinopathy effectively [40]. In this paper, we have proposed DRDiag, a Graph-ware deep learning multimodel for diabetic retinopathy detection. The experimental results on the APTOS2019 and MESSIDOR-2 datasets demonstrate significant improvements compared to other state-of-the-art methods, highlighting the effectiveness of our DRDiag model. A key limitation of our study is that it considers only one risk factor (the duration of diabetes, which is recognized as a primary risk factor) without accounting for other influential factors such as age, gender, dyslipidemia, and smoking. These factors have been shown to significantly impact the progression and severity of diabetic retinopathy. In the future, we envisage to incorporate these risk factors alongside blood pressure and glycated hemoglobin analysis to improve the classification performance of DR images.

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